

# Anshul Panigrahi

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## EDUCATION

### Vellore Institute of Technology

BTech in Information Technology; CGPA: 8.74/10

– Coursework: Machine Learning, Cloud Architecture, Computer Networks, Database Management Systems

Vellore, Tamil Nadu

Aug. 2023 – Jul. 2027

## EXPERIENCE

### Brandworks Technologies Private Limited

AI/ML Intern

Bengaluru, India

May 2026 – Jul. 2026

- Engineered a nine-stage data pipeline covering source audits, label normalization, perceptual-hash leakage removal, annotation filtering, and class balancing for 26,419 images across 8 categories.
- Trained and evaluated an 11M-parameter YOLOv8s object detector for 120 epochs, achieving 80.4% test precision, 67.6% recall, 66.2% mAP@50, and 51.6% mAP@50–95.

## TECHNICAL SKILLS

**Programming:** Python, TypeScript, JavaScript, SQL, C++, HTML, CSS

**Backend and Data:** FastAPI, Express, REST APIs, WebSocket, Kafka, Spark, Cassandra, DuckDB, SQLite, MongoDB, Redis

**Frontend and Platforms:** React, Next.js, Tauri, Docker, Git, Linux, Raspberry Pi

**AI & GenAI:** LLMs, RAG, AI agents, embeddings, vector search, workflow automation, PyTorch, TensorFlow/Keras, scikit-learn, Ollama, model training and evaluation

## PROJECTS

### Real-Time Predictive Maintenance | Python, Kafka, Spark, Cassandra, PyTorch, Next.js

- Architected a Lambda-architecture platform for 5,000 simulated factory machines, ingesting approximately 100 sensor messages per second through Kafka and Spark Structured Streaming into Cassandra and HDFS.
- Implemented streaming anomaly detection, batch prediction, graph-based cascade analysis, constrained mitigation, REST/SSE APIs, and a Next.js/Three.js dashboard rendering 5,000 machine states at 60 FPS.

### Eve Windows Dictation | Electron, Svelte, TypeScript, Python, WebSocket, SQLite, Faster-Whisper

- Built and maintained a local-first Windows dictation app with global hotkeys, a click-through live overlay, searchable local history, automatic clipboard insertion, and packaged speech-model workflows.
- Hardened the Electron-to-Python WebSocket transcription path across partial/final text, CPU/CUDA fallback and diagnostics, first-use model downloads, rapid session transitions, and NSIS release packaging; contributed 100+ attributed commits.

### ZRead | React, TypeScript, Tauri, Express, Gemini API, JSZip

- Developed a desktop-first EPUB reader with 2 storage adapters for local and browser use, supporting paginated reading, chapter navigation, highlights, saved vocabulary, configurable typography, and persisted progress.
- Integrated resilient OPF, spine, XHTML, and image parsing with 3 Gemini-backed actions—define, explain, and summarize—returning structured responses through an Express API.

### Stella | Python, DuckDB, FastAPI, React, WebSocket, Ollama, Docker

- Created a local-first analytics platform ingesting 6 wearable-data sources—Fitbit, Apple Health, Google Takeout, Oura, Garmin, and CSV—and normalized events into DuckDB for anomaly windows and lagged correlations.
- Delivered authenticated APIs, React analytics, PDF reports, and WebSocket chat with optional local-LLM summaries, a metrics-only fallback, and test coverage across 3 layers using Pytest, Vitest, and Playwright.

## ACHIEVEMENTS

### Published Patents | Indian Patent Office

2026

- System and Method for Zero-Trust Physical Enclave Security Using Continuous Biometric Entropy and RF-Based Perimeter Monitoring | IN202641067209 A1 | 5 Jun. 2026
- Medication Dispenser with On-Device User Authentication and Medication Verification | IN202641027860 A1 | 20 Mar. 2026
- Co-invented with a 5-member team: System and Method for Cognitive Load-Aware Distributed Task Scheduling using Hardware-Bound Processing Control and Predictive Task Migration | IN202641088096 A1 | 24 Jul. 2026

### Hackathons | IEEE CS and Yantra Central

2025

- **2nd Place** – IEEE CS Hackathon, SpineGuard posture-correction system, Sep. 2025
- **Finalist** – Yantra Central Hackathon, fingerprint spoofing detection, Feb. 2025

## CERTIFICATIONS

IBM Career Education Program – Building Agents with Agentic AI  
NVIDIA DLI Getting Started with Deep Learning  
Oracle Cloud Infrastructure 2025 AI Foundations Associate

Jul. 2026

Mar. 2026

Mar. 2026